Status	Finished
Started	Monday, 19 October 2026, 11:20 AM
Completed	Monday, 19 October 2026, 11:24 AM
Duration	3 mins 34 secs
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Quick sort
- c. Selection sort
- d. Heap sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Merging two datasets
- c. Removing duplicates from a list
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It is less efficient than bubble sort
- c. It sorts data using a single pass
- d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **5**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Quick Sort
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It sorts data out of the box efficiently
- c. It requires external libraries
- d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 0.00 out of 1.00

What is Bubble Sort known for?

- a. Being the most efficient sorting algorithm
- b. Bubbling up the largest element to its correct position with each pass
- c. Using the divide-and-conquer approach
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **9**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Insertion
c. Selection
d. Extraction

The correct answer is:
Insertion

Question **10**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Bubble Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By breaking the input into smaller parts and merging them
c. By comparing elements sequentially
d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Insertion sort

d. Bubble sort

The correct answer is:
Insertion sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems

b. It uses the heap data structure

c. It bubbles up the largest element in each pass

d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **14**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above

b. Selection

c. Searching

d. Duplicates

The correct answer is: All of the above

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems